

Mathematics A

Unit 1 • Chapter OL-T56 Classified

Cross-session • chapter-organised candidate

17 classified items • 53 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Unit 1 • Chapter OL-T56 • 17 items • 53 marks

Chapter	Items	Marks	Starts
01. OL-T56 Probability Toolkit	17	53	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Probability Toolkit

Unit 1 • 53 marks • 17 item(s)

June 2025 | 4MA1-2H Q11 | 3 marks

Recover population composition from a single-event probability

June 2024 | 4MA1-1H Q2 | 4 marks

Recover population composition from a single-event probability

November 2023 | 4MA1-1H Q4 | 2 marks

Complete a biased probability table using total one

November 2023 | 4MA1-1H Q4 | 2 marks

Calculate expected frequency from probability and trials

June 2023 | 4MA1-2H Q2 | 3 marks

Calculate expected frequency from probability and trials

November 2024 | 4MA1-2H Q5 | 4 marks

Calculate expected frequency from probability and trials

June 2022 | 4MA1-1H Q2 | 4 marks

Reverse expected frequency to find trials or probability

January 2022 | 4MA1-2H Q6 | 5 marks

Use category ratios in a probability model

January 2020 | 4MA1-1H Q4 | 2 marks

Use favourable outcomes over total outcomes

January 2020 | 4MA1-2H Q6 | 4 marks

Calculate expected frequency from probability and trials

June 2021 | 4MA1-1H Q6 | 3 marks

Complete a biased probability table using total one

June 2021 | 4MA1-1H Q6 | 2 marks

Calculate expected frequency from probability and trials

November 2021 | 4MA1-1H Q3 | 2 marks

Recover population composition from a single-event probability

November 2021 | 4MA1-1H Q3 | 2 marks

Recover population composition from a single-event probability

May-Nov 2020 | 1H Q2 | 4 marks

Recover population composition from a single-event probability

... 2 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 18

4MA1-2H Q11

UNIT 1
3 marks

11 Khalid has a box of counters.

17 of the counters are red
28 of the counters are blue
the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

(Total for Question 11 is 3 marks)

WORKING SPACE

Question 01

4MA1-1H Q2

MODULAR UNIT 1

4 marks

- 2 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

WORKING SPACE

Question 06

4MA1-1H Q4 (a)

MODULAR UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

- (a) Work out the value of x

$x = \dots\dots\dots$
(2)

WORKING SPACE

Question 07

4MA1-1H Q4 (b)

MODULAR UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

.....
(2)

WORKING SPACE

Question 13

4MA1-2H Q2

MODULAR UNIT 1

3 marks

- 2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	green	yellow	brown	pink
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

WORKING SPACE

Question 21

4MA1-2H Q5 M01

MODULAR UNIT 1

4 marks

- 5 Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

Number	1	2	3	4	5
Probability		0.14	0.17		0.21

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

.....

WORKING SPACE

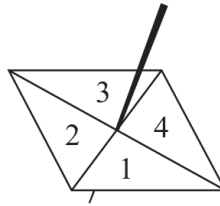
Question 01

4MA1-1H Q2 Q2

MODULAR UNIT 1

4 marks

- 2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

Working space for Question 01

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 23

4MA1-2H Q6

MODULAR UNIT 1

5 marks

- 6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio

number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

WORKING SPACE

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

One of the 40 babies is going to be chosen at random.

- (c) Find the probability that this baby has a weight of more than 5 kg.

(2)

YOUR WORKING

- 6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

Colour	red	blue	green	yellow
Probability		0.24		0.31

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.

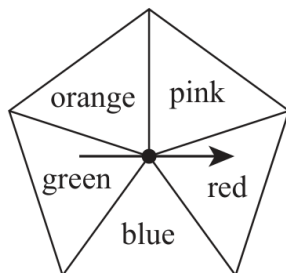
She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

YOUR WORKING

6 Grace has a biased 5-sided spinner.



Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

Colour	Red	Blue	Green	Orange	Pink
Probability	0.20	0.12	0.08		

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

(a) Work out the probability that the spinner will land on orange.

(3)

YOUR WORKING

Grace spins the arrow on the spinner 150 times.

(b) Work out an estimate for the number of times the spinner lands on blue.

(2)

YOUR WORKING

- 3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....
(2)

Jeevan puts the fish he took out, back into the tank.

He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....
(2)

YOUR WORKING

- 3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....
(2)

Jeevan puts the fish he took out, back into the tank.

He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....
(2)

YOUR WORKING

- 2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

.....
(Total for Question 2 is 4 marks)

YOUR WORKING

- 7 Some members of a library were asked to name the type of book that they each liked to read the best.

One of the members is chosen at random.

The table shows information about the probability of the type of book that this member answered.

Type of book	comedy	romance	mystery	thriller
Probability	0.24	0.40	$3x$	x

48 members answered comedy books.

Work out how many of the members answered mystery books.

(Total for Question 7 is 4 marks)

YOUR WORKING

9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.

The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.

The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

(Total for Question 9 is 3 marks)

YOUR WORKING
